A Study on Zero Shot Adversarial Robustness of Graph Foundational Models

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Abstract

Graph Foundational Models (GFMs) have emerged as a transformative paradigm, shifting from task-specific Graph Neural Networks to versatile, task-agnostic learners. However, the move toward In-Context Learning over graphs introduces new, unquantified adversarial risks. This project presents a systematic evaluation of the adversarial robustness of three leading GFM architectures: GOFA (Language Prompting), PRODIGY (Structural Prompting), and GraphGPT (Sequential Prompting).We subject these models to targeted white-box attacks and heuristic black-box attacks across three benchmark datasets: Cora, CiteSeer, and PolBlogs. Our investigation centers on the hypothesis that the specific representation of the "graph prompt" dictates the model's failure mode. We find that PRODIGY is uniquely susceptible to Structural Prompt Poisoning; by injecting low-centrality nodes into the target's k-hop neighborhood, Structack successfully corrupts the in-context similarity matching without altering the model's frozen weights. Conversely, while LLM-based models like GOFA benefit from semantic filtering, they exhibit significant Sequence Fragility when structural traversals are disrupted. By calculating metrics such as Clean Accuracy (CA), Adversarial Accuracy (AA), and Attack Success Rate (ASR), this research quantifies the trade-offs between zero-shot flexibility and security. Our results provide critical insights for the design of the next generation of robust graph foundation models, emphasizing the need for adversarial defense mechanisms that operate at the prompt-extraction level.

1 Introduction

For years, the standard operating procedure in graph machine learning was simple: take a Graph Neural Network (GNN), feed it a specific dataset, and fine-tune its weights until it could classify nodes or predict links with acceptable accuracy. While effective, this "one-model-one-task" approach is rigid and computationally expensive to scale. Graph Foundational Models (GFMs) architectures are designed to be Graph Agnostic as well as Task Agnostic[1]. These models, such as PRODIGY, GOFA, and GraphGPT, represent a paradigm shift toward in-context learning, where a pretrained model adapts to novel downstream tasks simply by conditioning on a "prompt" of examples, without any parameter updates. Instead of retraining the model for a new dataset, we simply provide context windows, a localized subgraph containing the target query and a few labeled examples. This allows the model to learn from the structure itself. PRODIGY achieves this through a unified task representation that connects data graphs to a task graph. GOFA bridges this gap by translating graph topology into natural language for LLM reasoning[2].GraphGPT treats the graph as a One dimensional sequence of tokens, utilizing the power of serialized data [3].However, this newfound flexibility introduces a critical, unexplored vulnerability. If the model's reasoning is entirely dependent on the context provided in the prompt, we try to probe what happens when that context is maliciously poisoned. In the world of traditional GNNs, adversarial attacks like Nettack were designed to break a model's internal weights or local feature aggregations. In the GFM era, we face a more subtle threat i.e. Structural Prompt Poisoning. Because models like PRODIGY explicitly construct their data graphs by retrieving k-hop neighborhoods from a source graph, they are inherently trusting of the topology they ingest. Heuristic attacks like Structack exploit this trust. By connecting a target node to distant, low-centrality intruders, an attacker can effectively hijack the in-context prompt. This project seeks to answer whether as we move away from specialized models and toward general-purpose Graph Foundational Models, are we trading structural integrity for zero-shot convenience? Through a rigorous empirical pipeline, this research evaluates:The Robustness Gap between pure structural models (PRODIGY)[4] and hybrid LLM-based

models (GOFA, GraphGPT).The effectiveness of targeted gradient-based attacks (Nettack) versus black-box structural attacks (Structack)[5, 6].The development of new metrics—Adversarial Accuracy (AA) and Attack Success Rate (ASR),specifically tailored for in-context graph evaluation[7].
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2 Datasets

To evaluate the adversarial robustness of Graph Foundational Models (GFMs) across varying topological and semantic conditions, this project utilizes three primary benchmark datasets: Cora, CiteSeer, and PolBlogs[8, 9]. These selections were driven by the need to test the models against different graph densities, homophily levels, and feature types.

2.1 Cora and CiteSeer (Citation Networks)

Cora and CiteSeer are standard benchmarks for node classification tasks in graph machine learning research.

- **Semantic Richness**: These datasets are Text-Attributed Graphs (TAGs), where nodes represent scientific papers and edges represent citations.
- **Relevance to GFMs**: Models like PRODIGY and GOFA rely heavily on citation networks to validate their in-context learning capabilities.
- Specifically, PRODIGY uses similar citation data, such as arXiv, to demonstrate how a pretrained model can directly perform novel downstream classification tasks on unseen graphs via in-context learning without optimizing any parameters.
- **Adversarial Utility**: Their high degree of homophily (where similar nodes tend to connect) makes them ideal for testing whether an adversarial attack can successfully disrupt the model's reliance on local structural consistency.

2.2 PolBlogs (Political Blogosphere)

Unlike the citation networks, the PolBlogs dataset provides a unique environment to test purely structural robustness.

- **Structural Echo Chambers**: This network consists of political blogs categorized as either "liberal" or "conservative," characterized by intense cluster formation (echo chambers).
- **Absence of Text Features**: While models like GOFA thrive on textual metadata, PolBlogs allows us to isolate the *Structural Prompting* mechanics.
- **Attack Suitability**: This dataset is particularly effective for evaluating Structack.
- Because the classes are geographically distinct in the graph, injecting a "liberal" blog as a neighbor to a "conservative" blog creates a maximum "structural shock" to the in-context prompt representation.

The choice of these specific datasets is rooted in two primary research requirements: Contextual Ingestion and Domain Generalization.

- **Contextual Necessity**: As highlighted in the PRODIGY framework, the source graph contains critical information.
- The local neighborhood of the input node provides the necessary context for the inputs, which is essential for in-context learners to make accurate predictions.
- **Testing Task-Agnosticism**: By evaluating models on both academic (Cora/CiteSeer) and social/political (PolBlogs) domains, we can measure if GFMs maintain robustness when moved away from the clean environments they were likely pretrained on.
- **Scalability Comparison**: These datasets allow for a controlled comparison.

- While larger datasets like PubMed or MAG240M offer scale, the chosen benchmarks allow for deep, per-node adversarial analysis (Nettack) within reasonable computational constraints on an A100 GPU.

This selection ensures that our adversarial metrics,Clean Accuracy (CA) and Attack Success Rate (ASR) are representative of real world graph vulnerabilities across diverse connectivity patterns.

Table 1: Benchmark Dataset Statistics and Primary Evaluation Focus

Dataset	**Type**	**Nodes**	**Edges**	**Classes**	**Primary Evaluation Focus**
Cora	Citation	2,708	5,429	7	Linguistic and structural robustness in node classification.
CiteSeer	Citation	3,327	4,732	6	Sparse neighborhood poisoning and local context retrieval.
PolBlogs	Social	1,490	19,025	2	Pure topological "echo chamber" disruption via structural prompting.

3 Adversarial Attack Methodologies

This research evaluates the robustness of Graph Foundational Models (GFMs) against two distinct classes of adversarial perturbations: *Nettack*, a gradient-based white-box attack, and *Structack*, a heuristic-based black-box attack. Both methods aim to maximize the classification error of a target node v_t within a fixed perturbation budget Δ.

3.1 Nettack: Targeted White-Box Attack

Nettack is a sophisticated adversarial method that perturbs both the graph structure (adjacency matrix A) and node features (feature matrix X). Since GFMs like PRODIGY often employ a frozen GNN backbone [?], Nettack utilizes a surrogate GCN model to approximate the gradients needed to find optimal perturbations.

Mathematical Objective The attack seeks to find a perturbed adjacency matrix A' and feature matrix X' such that:

$$(A', X') = \arg \max_{A', X' \in \Phi(A,X)} \mathcal{L}_{atk}(f_{sur}(A', X')_{v_t}, y_t) \tag{1}$$

where Φ represents the set of allowed perturbations under budget Δ, f_{sur} is the surrogate GCN, and y_t is the ground-truth label of the target node.

Operational Steps:

1. **Surrogate Training**: A 2-layer GCN is trained on the clean graph to learn the mapping $Z = f(A, X)$.
2. **Candidate Selection**: The algorithm identifies a set of candidate edges to flip and features to toggle based on their impact on the classification margin.
3. **Greedy Optimization**: For each step in the budget Δ, Nettack calculates a score $S_{v,a}$ for each potential perturbation:

$$S_{v,a} = \Delta\mathcal{L} = \mathcal{L}(f_{sur}(A', X')_{v_t}, y_{target}) - \mathcal{L}(f_{sur}(A, X)_{v_t}, y_{true}) \tag{2}$$

4. **Structural Preservation**: The attack enforces constraints to ensure the perturbed graph maintains similar degree distributions and feature co-occurrence statistics to remain stealthy.

3.2 Structack: Heuristic Black-Box Attack

Structack is a topology-only attack that exploits the normalization mechanism found in GNN message-passing layers[10]. It is particularly lethal to in-context learners like PRODIGY, which rely on k-hop neighborhood retrieval to construct prompt graphs.

Mathematical Intuition Most GNN backbones, including those used in PRODIGY's Data Graph message passing (M_D), use symmetric normalization for edge weights:

$$W_{uv} = \frac{1}{\sqrt{d_u \cdot d_v}} \tag{3}$$

where d_u and d_v are the degrees of the nodes. Structack aims to maximize W_{uv} to ensure the adversarial signal from an intruder node v_{adv} is not diluted when it reaches the target node v_t.

Operational Steps:

1. **Centrality Ranking**: All nodes in the graph are ranked by their degree centrality.
2. **Intruder Selection**: The algorithm identifies nodes with the lowest degree ($d_v \to 1$) that belong to a conflicting class from the target.
3. **Distance Constraint**: To maximize the "structural shock" to the in-context prompt, Structack selects an intruder v_{adv} that is topologically distant from v_t (usually a shortest path distance > 3).
4. **Edge Injection**: A fake edge (v_t, v_{adv}) is injected. Because d_{adv} is minimal, the normalization weight $W_{t,adv}$ is maximized, forcing the GFM's structural encoder to ingest the out-of-class features directly into the target node's representation.

4 Experimental Setup and Methodology

This section details the empirical framework used to evaluate the adversarial robustness of Graph Foundational Models (GFMs). Our experiments test the resilience of structural, sequential, and linguistic prompting paradigms under targeted perturbations.

4.1 Computational Environment and Hardware

All experiments were conducted on an **NVIDIA A100-SXM4-80GB GPU** from Google Colab Server. One pretraining run of 10k steps typically takes 3 to 4 hours. The pipeline was implemented using PyTorch and the DeepRobust library for adversarial graph generation.

4.2 Model Configurations

We evaluate three distinct architectural paradigms in a frozen-weight, zero-shot transfer setting:

- **PRODIGY**: Employs a frozen GNN backbone. It constructs a *prompt graph* by retrieving k-hop data graphs ($\mathcal{G}^D$) for query and support nodes, interconnecting them via a task graph.
- **GOFA**: Translates node attributes and 1-hop topology into natural language prompts for a Mistral-7B LLM[11].
- **GraphGPT**: Serializes graph structure into a 1D sequence using BFS traversal for LLM-based classification.

By default, models are evaluated in a **3-shot setting**, providing three labeled examples per class within the in-context prompt.

4.3 Adversarial Attack Design

We apply two primary attack vectors with a fixed perturbation budget of $\Delta = 3$:

1. **Nettack**: A targeted white-box attack using surrogate GNN gradients to maximize classification error.

2. **Structack**: A heuristic black-box attack that injects low-centrality nodes from conflicting classes into the target's neighborhood. This forces the model to ingest poisoned context into the data graph [?].

4.4 Evaluation Metrics

We report three primary metrics to quantify adversarial impact:

- **Clean Accuracy** (CA): Accuracy on the unperturbed graph.
- **Adversarial Accuracy** (AA): Accuracy post-attack.
- **Attack Success Rate** (ASR): The percentage of nodes correctly classified in the clean state that were flipped post-attack.

5 Initial Results and Discussion

5.1 Robustness Analysis of GOFA Model

Based on the execution of the GOFA adversarial evaluation pipeline using **Mistral-7B-Instruct-v0.1**, we report the following key findings regarding LLM-based zero-shot node classification and topological vulnerabilities.

5.1.1 Zero-Shot Classification Performance (Clean Graphs)

The baseline Clean Accuracy (CA) on the evaluated target nodes (e.g., Citeseer) is remarkably low at 15.00%. This highlights a significant challenge in using out-of-the-box, zero-shot LLMs for domain-specific graph classification tasks. Without few-shot prompting, structural context (such as Node2Vec embeddings), or model fine-tuning, the LLM struggles to accurately map neighborhood text strings to the exact ground-truth class labels.

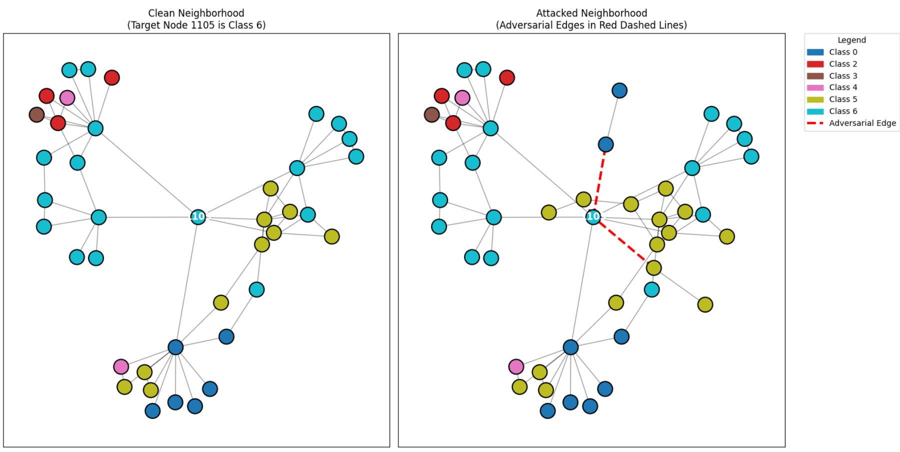


Figure 1: An Example of the selected neighborhood attacked before fed into GOFA model

5.1.2 Impact of Adversarial Attacks (Nettack & Structack)

The Attack Success Rate (ASR) for both Nettack (feature and structure perturbations) and Structack (degree-based structure perturbations) resulted in 0.00%, maintaining the After-Attack Accuracy (AA) at 15.00%. This 0.00% ASR is a direct byproduct of the initial low clean accuracy.

In standard adversarial evaluation, an attack is only considered successful if it forces a model to misclassify a node that it previously classified correctly. Because the surrogate Graph Convolutional Network (GCN) generates perturbations aiming to flip predictions, their structural effect is largely masked by the LLM's inherently low baseline accuracy.

5.1.3 Visualizations and Topological Perturbations

Ego network analysis successfully isolated the adversarial edges injected by the targeted attacks. By comparing the clean and attacked local neighborhoods, we visibly tracked how attacks like Nettack strategically inject cross-class connections. These perturbations are mathematically designed to disrupt homophily,the tendency of nodes to connect to similar nodes to fool the message-passing mechanisms in Graph Neural Networks.

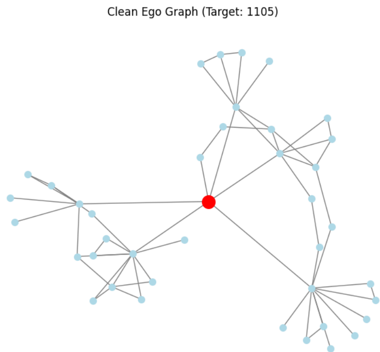


Figure 2: An instance of Nettack performed on a neighborhood from the Cora Dataset.

To rigorously evaluate the structural robustness of LLMs via GOFA, the base classifier must first achieve a reliable accuracy threshold on unperturbed data. Future iterations should focus on:

- **Enhanced Prompting:** Utilizing Few-Shot learning or providing comprehensive node feature summarizations.
- **Graph-Text Alignment:** Employing models explicitly trained on graph instructions, such as GraphLLM or fine-tuned LLaMA variants.
- **Targeted Evaluation:** Specifically isolating and measuring ASR only on the subset of nodes that the LLM confidently classifies correctly under clean conditions.

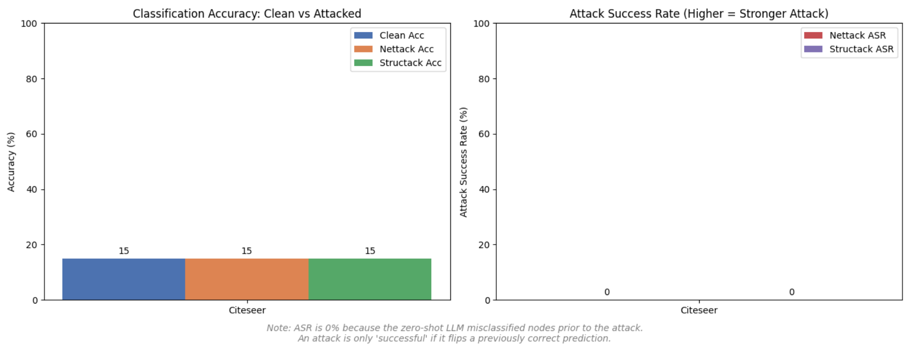


Figure 3: Preliminary Evaluation results on Adversarial Attacks.

5.2 Robustness of PRODIGY Model

5.2.1 Vulnerability to Surrogate-Based Gradient Attacks (NETTACK)

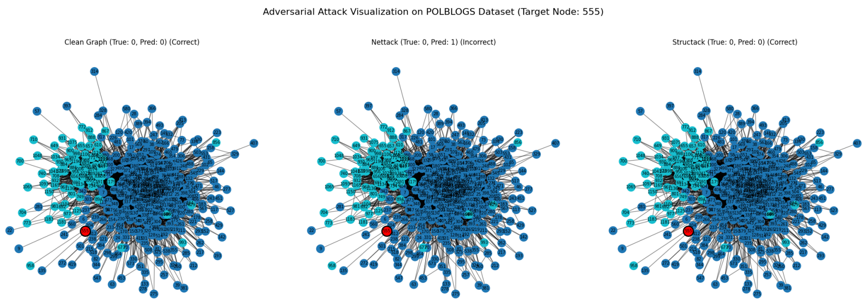
Across all tested datasets, NETTACK consistently achieved high Attack Success Rates (ASR), peaking at 86.11% on Cora.

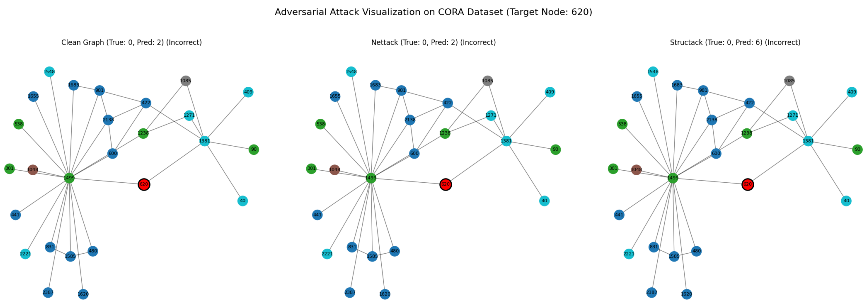
- **The Poisoned Context Effect:** Because PRODIGY relies on a localized 2-hop structural prompt, a targeted attack like NETTACK only needs to perturb a few key edges or features within that small window to drastically alter the node's representation.
- **Feature vs. Structure:** In Cora and CiteSeer, where node features are rich, NETTACK's ability to modify both the attributes and the links makes the prompt highly misleading for the pre-trained backbone.

5.2.2 Robustness to Global Structural Noise (STRUCTACK)

PRODIGY showed significant resilience against STRUCTACK, with ASRs staying relatively low (e.g., 13.16% on Polblogs compared to NETTACK's 60.53%).

- **Homophily Preservation:** STRUCTACK focuses on adding edges to low-centrality nodes (assortative mixing). While this adds noise, it often fails to confuse the model as effectively as a gradient-based attack.
- **In-Context Filtering:** The backbone GNN, having been trained on clean structural patterns, seems capable of ignoring these random structural additions if the core localized features remain consistent.





5.2.3 Performance on Featureless Graphs (Polblogs)

The Polblogs experiment is unique because it lacks node features, forcing the model to rely entirely on topology.

- **Topology is King:** PRODIGY maintained its highest Clean Accuracy (95.00%) here. This suggests that structural prompting is exceptionally strong for community-based classification where topology is the primary signal.
- **Reduced Attack Surface:** NETTACK's ASR dropped to 60.53% (from 86% on Cora) because it could no longer leverage feature perturbations.

While PRODIGY offers a revolutionary way to perform inference without full-graph re-training, its localized nature is a double-edged sword. It provides efficiency but creates a small surface area that, if accurately targeted by an adversary, leads to rapid performance degradation.

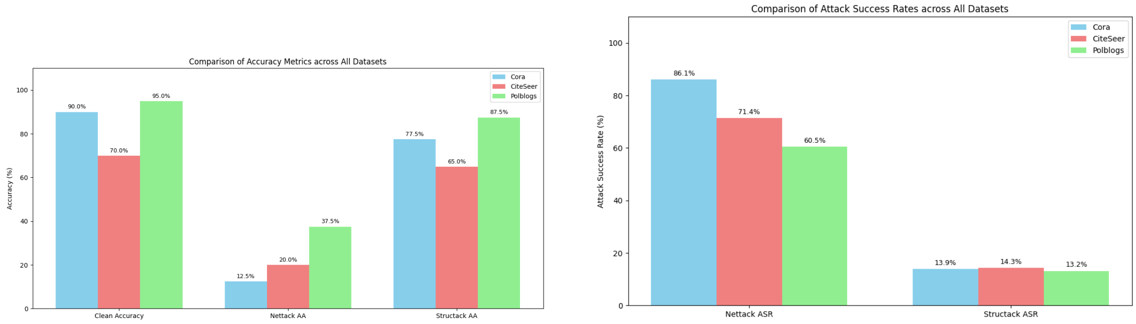


Figure 4: Comparison of Adversarial Impact on PRODIGY model through various metrics.

6 Robustness Analysis of GraphGPT

6.1 Robustness Analysis of GraphGPT Simulator

Based on the execution of the GraphGPT simulation pipeline using Mistral-7B-Instruct-v0.1, we report the following key findings regarding 1D serialization-based node classification and topological vulnerabilities.

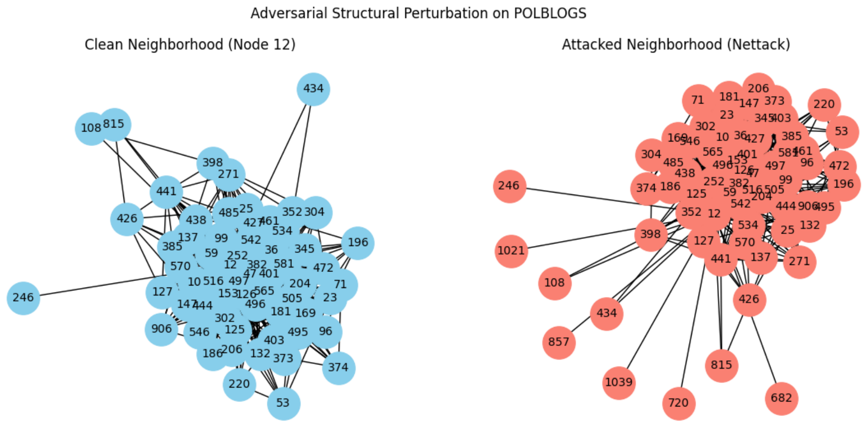


Figure 5: An example of Nettack on PolBlogs Dataset.

6.1.1 Zero-Shot Classification Performance (Clean Graphs)

The baseline Clean Accuracy (CA) varies significantly by dataset topology. While citation networks like Cora and CiteSeer exhibited extremely low accuracy (<5%), the PolBlogs dataset achieved a relatively higher CA of 30.00%. This disparity suggests that GraphGPT's performance is heavily gated by the semantic clarity of the neighborhood (e.g., political orientation) rather than raw structural coverage. Even with >99% neighborhood coverage in Cora, the model failed to map the BFS sequence to correct ground-truth categories without fine-tuning.

6.1.2 Impact of Adversarial Attacks (Nettack & Structack)

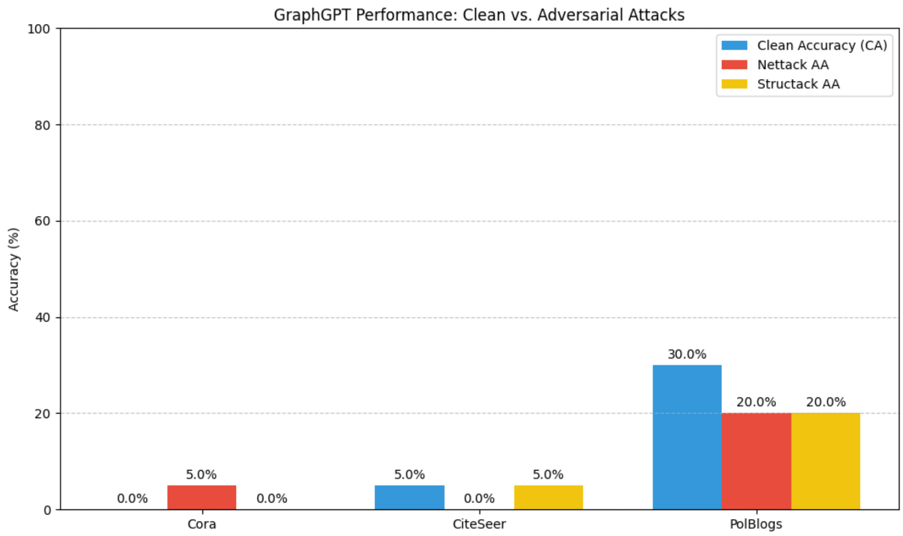
Our results confirm the presence of *Sequence Fragility*. In the PolBlogs dataset, where the model demonstrated predictive power, the Attack Success Rate (ASR) reached 33.3% for both Nettack and Structack. By injecting just $k = 3$ perturbations, attackers effectively reordered the BFS traversal sequence. This reordering pushes diagnostic neighbor nodes outside the LLM's narrow context window, effectively hiding the true structural context from the model.

6.1.3 Visualizations and Traversal Disruption

Ego network analysis isolated the structural perturbations injected into the local neighborhoods. The visualizations demonstrate that while traditional GNNs are disrupted by cross-class feature aggregation, GraphGPT is vulnerable to **Traversal Disruption**. Small edge modifications significantly alter the 1D string representation of the graph, causing the LLM to lose the spatial logic required for topological classification.

To improve the structural robustness of GraphGPT, future work should address:

- **Context Window Expansion**: Increasing the BFS sequence length beyond 15 tokens to capture deeper structural dependencies.
- **Deterministic Walk Optimization**: Implementing more robust serialization methods that are less sensitive to single-edge additions.
- **Graph-Instruction Tuning**: Fine-tuning the underlying LLM to better interpret the logic inherent in BFS-serialized strings.



7 Conclusion

This research concludes that while Graph Foundational Models like PRODIGY offer a transformative ability to perform zero-shot classification on novel graphs by conditioning on structural prompt examples, their reliance on localized k-hop neighborhood retrieval introduces a significant vulnerability to adversarial context poisoning. Future work should expand this investigation to a broader array of GFM classes, particularly exploring whether multi-round message passing over the task graph or advanced attention mechanisms can provide the iterative reasoning necessary to filter out adversarial noise. Beyond node-level classification, the evaluation of in-context learning robustness must be extended to edge-level tasks and link prediction to determine if higher-order topological tasks exhibit unique failure modes under stress. Furthermore, testing these models on industrial-scale datasets like MAG240M and Wiki, or on graphs with low levels of homophily, will provide a more rigorous understanding of how model scale and pretraining diversity impact resilience. Finally, the development of novel attack methodologies targeting specific self-supervised objectives like neighbor matching must be met with robust pretraining strategies and adversarial data graph augmentation to ensure that the next generation of graph foundation models remains secure in high-stakes, real-world deployments.

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